

ROTTNEST ISLAND, Australia

WMO: 946020

Lat: 32.0069S

Lon: 115.5022E

Elev: 141

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.9	52.2	37.4	32.9	57.1	39.0	35.1	57.0	41.8	60.4	38.5	59.9	14.9	90	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.4	88.6	68.1	84.6	67.5	81.3	67.1	73.7	78.8	71.8	77.5	70.5	76.5	12.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.2	119.6	76.0	70.2	111.7	74.4	68.6	105.8	73.8	37.4	78.9	35.7	77.5	34.6	76.4	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	34.8	31.7	29.0	47.9	98.0	1.3	3.0	46.9	100.1	46.1	101.9	45.4	103.6	44.4	105.8	
(5)				43.5	75.1	1.5	3.3	42.5	77.4	41.6	79.3	40.7	81.1	39.6	83.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	72.0	73.1	71.6	68.4	64.7	61.1	59.2	59.3	60.2	62.6	66.6	69.5
	DBStd	6.39	4.52	4.88	4.78	4.10	3.57	3.20	3.10	3.10	3.33	3.80	4.42	4.91
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	821	1	1	3	12	49	121	180	179	147	94	27	7
	CDD50	5712	682	646	670	551	454	333	286	288	307	392	498	605
	CDD65	1055	218	227	207	113	38	4	1	1	4	20	75	147
	CDH74	3929	876	1115	829	224	29	0	0	0	6	54	230	566
	CDH80	1023	236	306	206	34	0	0	0	0	1	10	64	166
Wind	WSAvg	16.7	18.7	17.9	16.9	14.8	14.5	15.8	16.4	15.7	16.0	16.7	17.9	19.0
Precipitation	PrecAvg	24.8	0.2	0.4	0.6	1.3	3.6	5.2	5.2	3.4	2.2	1.4	0.7	0.3
	PrecMax	34.5	1.3	2.6	2.5	3.1	7.2	10.4	11.1	7.8	4.5	3.6	2.4	1.5
	PrecMin	13.1	0.0	0.0	0.0	0.0	1.0	1.9	2.4	1.7	0.6	0.0	0.0	0.0
	PrecStd	5.1	0.3	0.6	0.7	0.9	1.9	2.1	1.7	1.2	0.9	0.8	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.9	93.3	92.4	84.4	77.3	70.8	69.1	70.1	73.2	80.5	87.3	91.3
		MCWB	68.3	69.8	69.2	65.0	62.6	59.9	60.6	60.8	59.8	62.1	64.0	66.8
	2%	DB	85.9	87.7	85.2	79.3	73.9	68.8	66.9	67.4	69.1	74.0	80.1	84.5
		MCWB	68.8	69.5	67.2	64.7	63.2	59.9	59.3	60.5	60.7	61.5	64.2	66.8
	5%	DB	81.4	83.4	81.0	76.1	71.6	67.4	65.3	65.7	66.9	70.3	75.3	79.8
		MCWB	68.2	68.9	66.6	65.0	62.5	59.3	58.3	59.0	59.9	62.0	63.6	66.2
	10%	DB	78.1	80.0	78.0	73.8	69.9	66.1	63.9	64.3	65.3	68.1	72.4	76.1
		MCWB	68.2	68.4	67.6	64.7	62.0	58.5	57.4	57.8	58.9	61.0	63.6	65.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.0	76.7	76.3	71.2	69.0	64.8	63.8	64.8	65.0	67.1	69.3	71.5
		MCDB	81.4	83.1	77.5	73.9	71.7	67.4	66.4	67.4	67.9	70.6	74.6	79.9
	2%	WB	72.8	74.4	72.4	69.8	66.5	62.9	61.7	62.5	63.1	65.1	67.5	69.9
		MCDB	78.3	79.6	77.1	73.3	70.5	66.5	65.1	65.6	66.8	69.2	73.1	77.5
	5%	WB	71.1	72.1	70.8	68.1	64.9	61.5	60.1	60.6	61.3	63.7	66.2	68.6
		MCDB	76.8	78.2	76.2	72.9	69.7	65.7	63.7	64.2	65.6	68.2	72.3	75.8
	10%	WB	69.7	70.5	69.5	66.6	63.2	60.0	58.6	59.0	59.6	62.3	64.9	67.3
		MCDB	75.9	77.2	75.3	72.2	68.8	64.7	62.7	63.2	64.3	67.3	71.0	74.2
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.4	10.7	9.2	8.4	7.6	7.4	8.0	8.5	9.5	10.3	10.4
		MCDBR	17.0	17.5	15.9	12.6	10.4	8.8	8.6	9.4	10.7	13.6	16.5	17.2
		MCWBR	8.3	8.3	8.3	7.6	7.0	6.4	6.5	6.8	7.4	8.4	9.0	8.2
	5% WB	MCDBR	13.0	13.5	11.2	10.0	8.8	8.0	7.7	8.1	9.7	11.0	12.8	13.5
MCWBR		7.9	7.6	7.8	7.1	7.6	7.6	7.3	7.5	7.9	7.5	8.1	7.1	
Clear-Sky Solar Irradiance	taub	0.339	0.339	0.325	0.318	0.309	0.303	0.302	0.309	0.326	0.326	0.327	0.328	
	taud	2.605	2.622	2.659	2.661	2.658	2.652	2.652	2.627	2.556	2.575	2.594	2.611	
	Ebn at Noon	317	311	305	291	278	272	278	290	299	312	319	322	
	Edh at Noon	33	31	28	25	23	21	22	26	30	32	33	33	
All-Sky Solar Radiation	RadAvg	2600	2344	1934	1408	1046	848	898	1180	1585	2069	2466	2666	
	RadStd	112	76	92	81	66	59	58	59	94	77	96	96	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)		N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page